

Technical Document #711: Retrieval Augmented Generation - Comprehensive Overview

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Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Re-ranking models score retrieved documents for relevance to the query.